RBI Inflation Analysis Using CPI and Repo Rate

1. Introduction

Inflation is a key macroeconomic indicator influencing monetary policy decisions. The Reserve Bank of India (RBI) uses the policy repo rate as a primary tool to control inflation. This study analyzes the relationship between inflation and repo rate using time series modeling.

2. Data Description

- CPI (Combined, General, All India)
- Policy Repo Rate
- Time Period: ~2012–2026
- Frequency: Monthly

3. Methodology

- Data cleaning and preprocessing
- Conversion to time series format
- Creation of lag features for repo rate and inflation
- Model building using:
  - Linear Regression
  - Random Forest

4. Results

4.1 Repo Rate Only Model

- $R^2 \approx 0.01$
- Conclusion: Repo rate alone does not explain inflation effectively

4.2 Lag-Based Model

- $R^2 \approx 0.84$
- Conclusion: Inflation exhibits strong persistence

4.3 Random Forest Model

- Captures non-linear relationships
- Better performance in capturing fluctuations

5. Key Insights

- Inflation responds to monetary policy with a lag
- Inflation is highly persistent (depends on past values)
- Repo rate has limited standalone impact
- External factors (oil prices, supply shocks) play significant roles

6. Conclusion

The study shows that while repo rate is an important policy tool, it cannot independently explain inflation dynamics. Incorporating lagged inflation significantly improves model performance, highlighting the importance of persistence in inflation behavior.

7. Future Work

- Include additional variables (oil prices, GDP, food inflation)
- Use advanced models (XGBoost, ARIMA)
- Policy simulation analysis